

C.AAA (Authentication, Authorization, Accounting): Three core functions, AAA in Networking & Security, AAA Deployment Models.

Task 1.

Write a short scenario where a user tries to access a private WiFi network (like at a campus or airport) and describe how each of the three AAA functions—Authentication, Authorization, and Accounting—would apply step by step in this situation.

1. Scenario: User Accessing a Private Campus Wi-Fi

Scenario:

A student named Rahul wants to connect his laptop to the private Wi-Fi network of his college. The college uses a AAA server to control who can access the network and what they are allowed to do.

Step 1 — Authentication

Rahul connects to the college Wi-Fi and enters his student username and password.

- The Wi-Fi access point sends Rahul's credentials to the AAA server.
- The AAA server checks whether the username and password are valid.
- If the credentials are correct, Rahul's identity is authenticated.
- If they are incorrect, access is denied.

Purpose: Authentication answers "Who are you?"

Step 2 — Authorization

After Rahul is authenticated, the AAA server determines what he is allowed to access.

For example:

- Rahul can access the college learning portal.
- He can browse approved websites.
- He cannot access administrative systems.
- He may receive a specific VLAN or network policy.

Purpose: Authorization answers "What are you allowed to do?"

Step 3 — Accounting

While Rahul uses the Wi-Fi, the AAA system records information about his network session.

It may record:

- Username
- Login time
- Logout time
- IP address
- Amount of data transferred
- Access attempts

Task 2.

Create a table comparing at least two different AAA deployment models (such as centralized vs distributed), listing three advantages and three disadvantages of each for a company like Zomato that handles sensitive user data.

Deployment Model	Advantages	Disadvantages
Centralized AAA	1. One central location for managing users and policies. 2. Easier monitoring and auditing. 3. Consistent security policies across the company.	1. Central server failure can affect many services. 2. Can become a performance bottleneck. 3. Requires reliable network connectivity to the AAA server.
Distributed AAA	1. AAA services can continue even if one server/location fails. 2. Better performance because requests can be handled locally. 3. Useful for companies with multiple branches or regions.	1. More expensive to deploy and maintain. 2. Policies must be synchronized between servers. 3. Troubleshooting and monitoring are more complex.

Example for Zomato

A company such as Zomato, which handles sensitive customer and business information, could use AAA to control access to internal applications, databases, cloud resources, and employee systems.

Centralized: One primary AAA infrastructure controls access throughout the organization.

Distributed: Multiple AAA servers can be deployed across different offices or regions and synchronized with each other.

Task 3.

Draw a simple network diagram (can be hand-drawn and photographed) showing how AAA servers are integrated into a typical enterprise network for user access control, labeling where Authentication, Authorization, and Accounting processes happen.

Authentication:

The AAA server verifies the employee/student username and password.

Authorization:

After successful authentication, the AAA server determines which network resources the user can access.

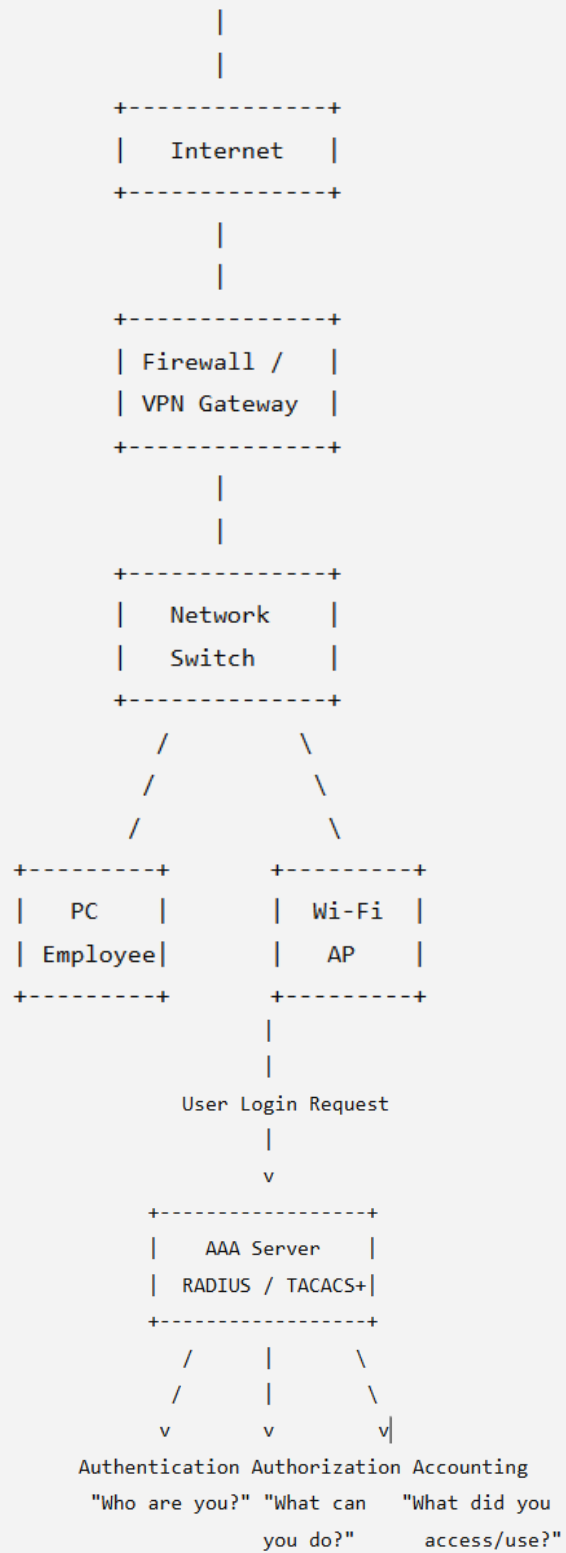
Accounting:

The AAA server records login/logout times, access attempts, IP addresses, and other session information.

Common AAA protocols

- RADIUS – commonly used for Wi-Fi and network access.
- TACACS+ – commonly used for administrative access to network devices.

ENTERPRISE NETWORK



Task 4.

**Pick any app you use daily (Instagram, Paytm, or Flipkart) and write a paragraph explaining how AAA concepts could be applied to secure user logins and actions within that app.

Hint: Think about what happens when you log in, what features you are allowed to use, and how your actions might be tracked for security or billing.**

- Instagram can use AAA concepts to protect user accounts and control user activities.
- Authentication happens when a user logs in using their username/password and possibly a verification code or other security mechanism.
- Authorization determines what the authenticated user is permitted to do, such as posting content, sending messages, changing account settings, or accessing account-management features.
- Accounting can record security and activity information such as login times, device information, IP addresses, and account actions.
- These records can help detect suspicious activity, investigate unauthorized access, and maintain security records

Task 5.

Use ChatGPT or any AI tool to generate a sample AAA policy for a college WiFi network that restricts access to students only, allows certain websites, and logs all access attempts. Paste the generated policy and briefly explain how each AAA function is addressed.

College Wi-Fi AAA Security Policy

Policy Name: College Student Wi-Fi AAA Access Policy

Objective:

To provide secure Wi-Fi access only to registered students and maintain records of network access.

Authentication Policy

1. Only students with valid college credentials can connect to the Wi-Fi.
2. Students must authenticate using their college username and password.
3. Invalid credentials will result in access being denied.
4. After repeated failed login attempts, the account may be temporarily blocked.

5. Guest users must use a separate guest network.

Authorization Policy

1. Authenticated students are assigned to the student Wi-Fi network/VLAN.
2. Students may access:
 - College website
 - Learning Management System (LMS)
 - Educational websites
 - Approved email and cloud services
3. Access to administrative systems and restricted internal resources is prohibited.
4. Websites categorized as malicious, illegal, or inappropriate can be blocked by the network firewall/content filter.
5. Different access permissions may be assigned to students, faculty, and administrators.

Accounting Policy

1. All successful and failed Wi-Fi login attempts must be logged.
2. The system should record:
 - Username
 - Date and time
 - IP address
 - Device information
 - Login/logout information
 - Data usage/session information where applicable
3. Logs should be protected from unauthorized modification.
4. Security administrators should periodically review logs for suspicious activity.
5. Logs should be retained according to the college's security and privacy requirements.

AAA Function	How the Policy Implements It
Authentication	Verifies that the person connecting is a registered student using college credentials.
Authorization	Determines which websites, VLANs, applications, and resources the student can access.
Accounting	Records login attempts, sessions, IP addresses, device information, and usage for security monitoring.